

Empirical Evaluation

Quantify the experience

Usability - Definition

ISO 9241-11: Guidance on Usability (1998)

Usability

The extent to which a product can be used by specified users to achieve specified goals with **effectiveness**, **efficiency** and **satisfaction** in a specified context of use.

- Effective
- Efficient
- Satisfactory

How to measure usability?

- Effectiveness
- Efficiency
- Satisfaction

Usability - How to measure it?

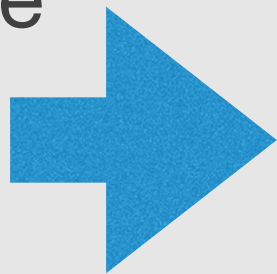
- Completion rates
- Usability Problems (UI)
- Task time
- Task level satisfaction
- Test level satisfaction
- Expectation
- Errors
- Page views/clicks
- Conversion
- Single Usability Metric (SUM)

Completion Rate

- Fundamental usability metric
- Advantages:
 - Easy to understand, engineers, managers etc
 - Popular - used by over 80% of evaluations
 - Easy to collect, at any stage of dev.
- How many users should I test?
 - Use a sample size calculator
 - 14% of users who fail a task still rate it as super easy!

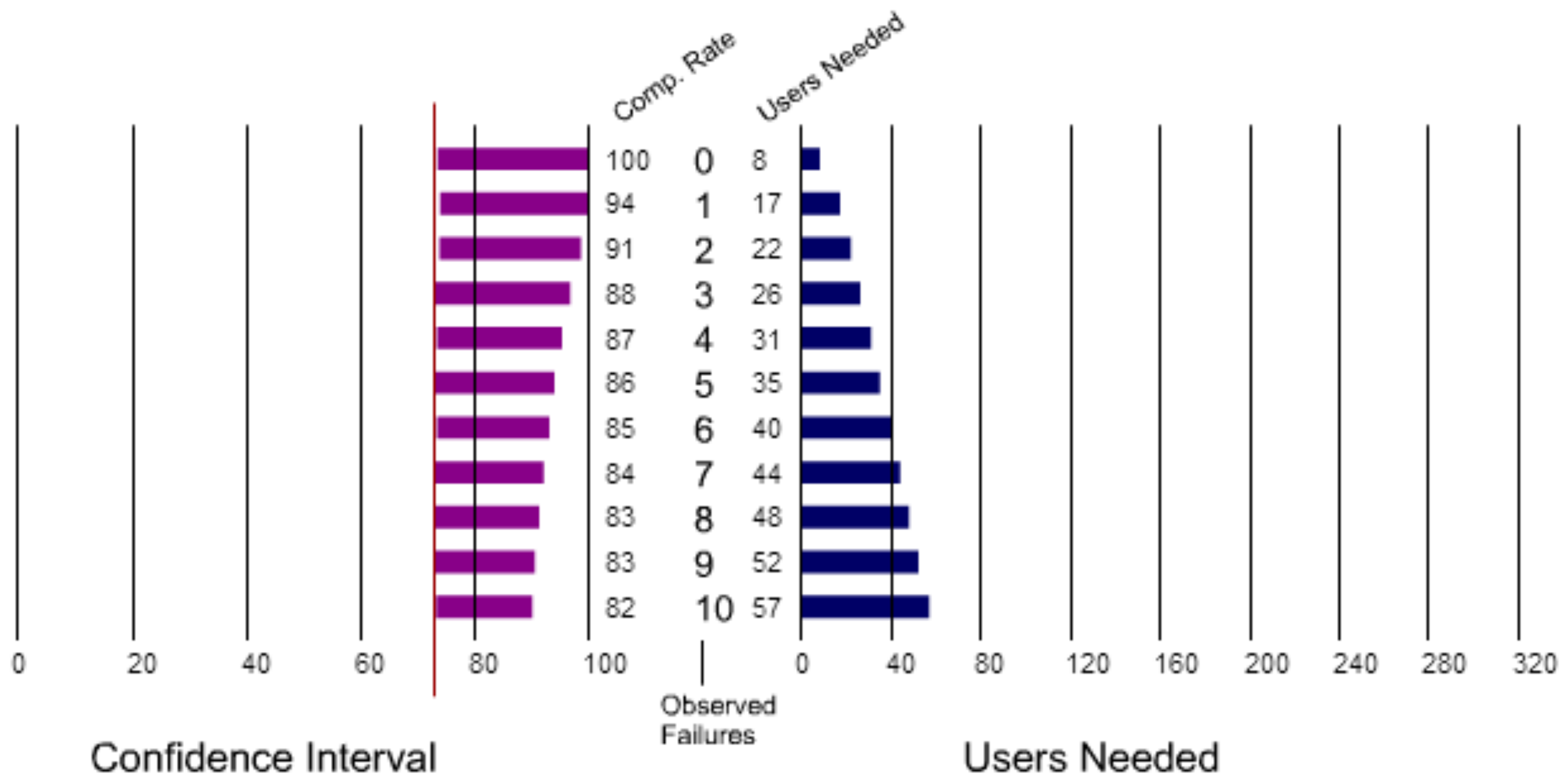
Completion Rate

- 100% completion rate
- OR
- 100% failure rate

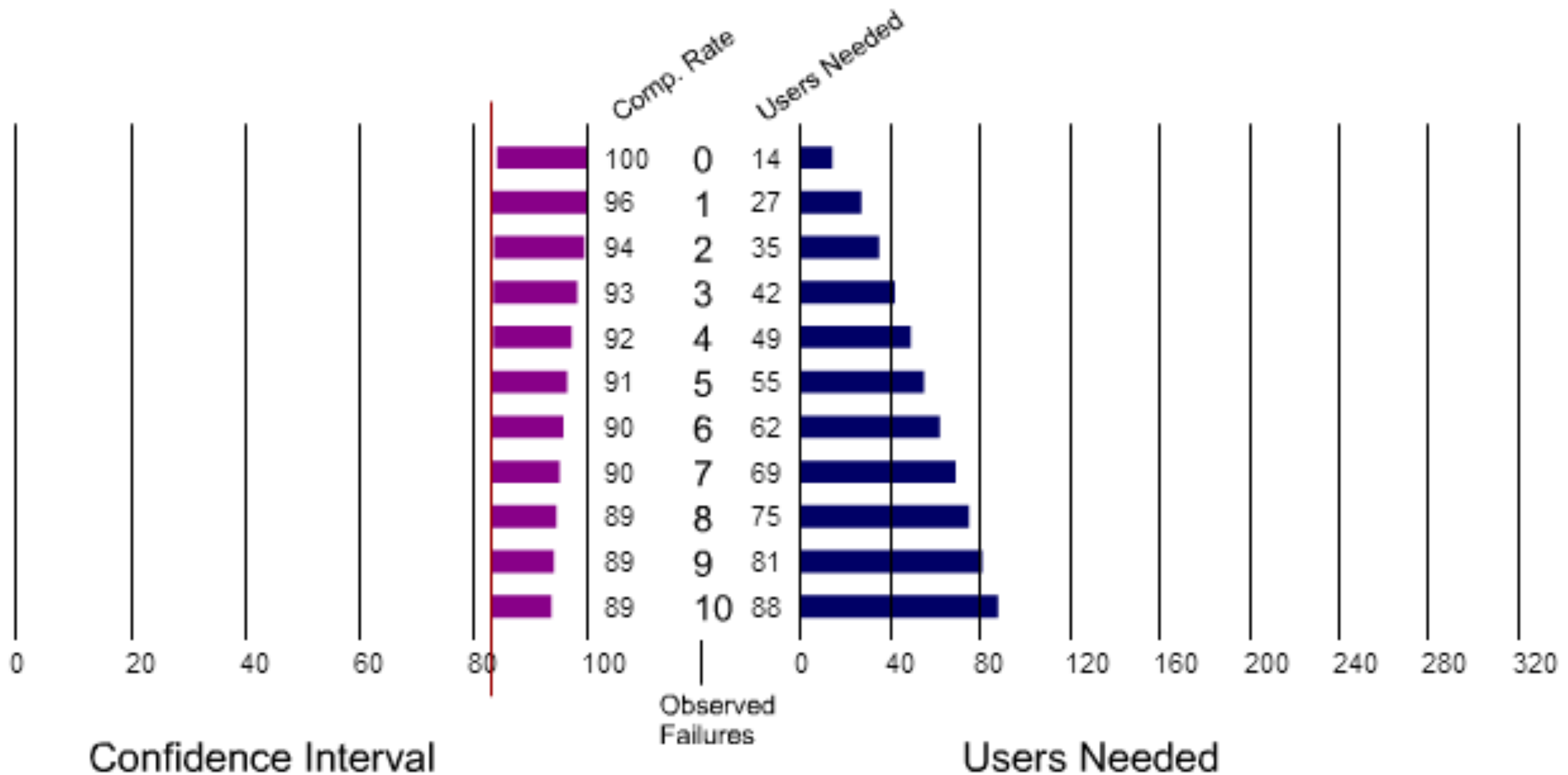


Not reliable for small groups of users

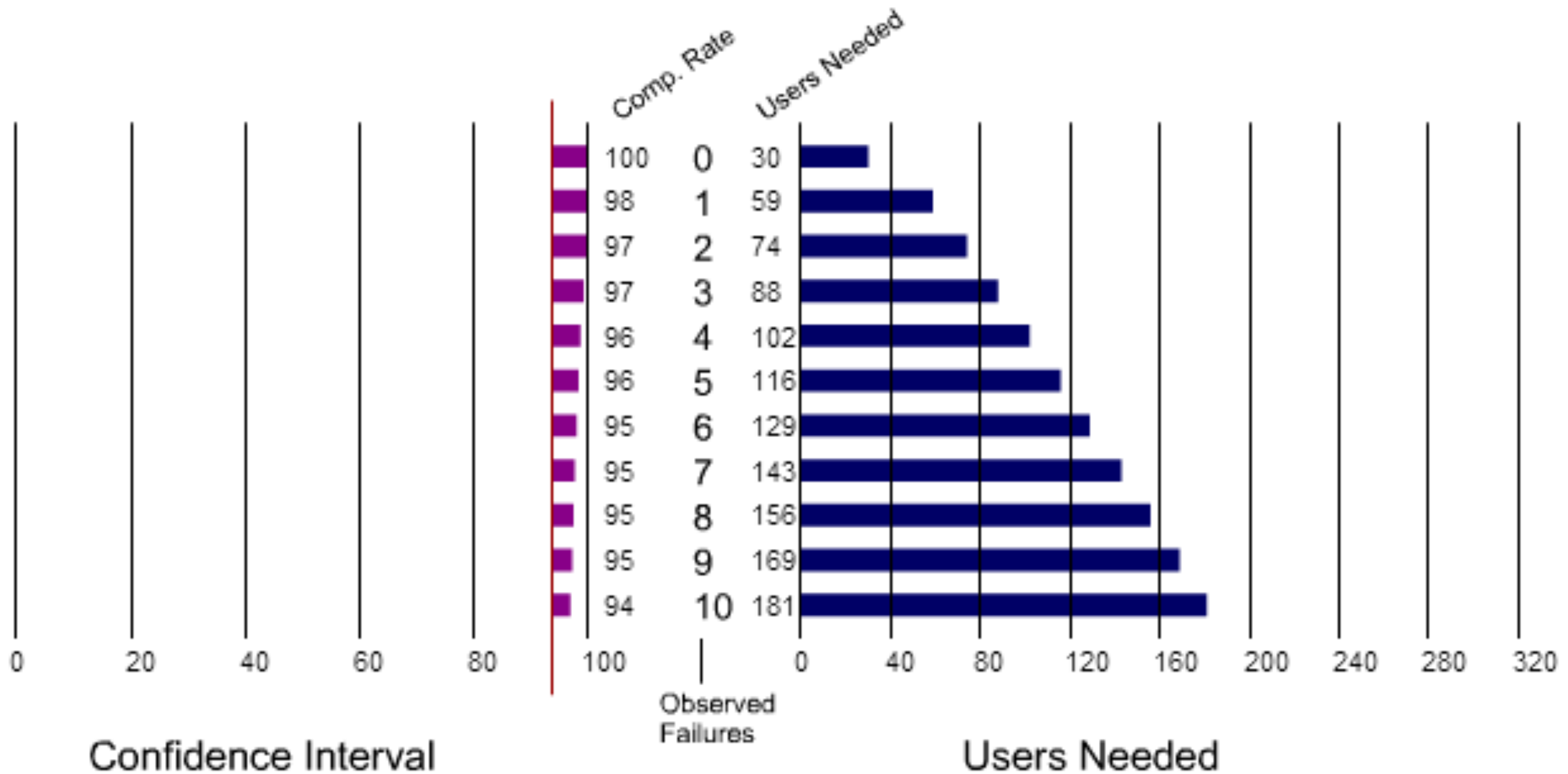
> 70% Confidence Factor



> 80% Confidence Factor



> 90% Confidence Factor



Usability Problems - Track

- Give it a name or id
- Description
- Give it a severity rating or assign it to a category
 - Prevents task completion
 - Causes delay or frustration
 - Minor effect on the task

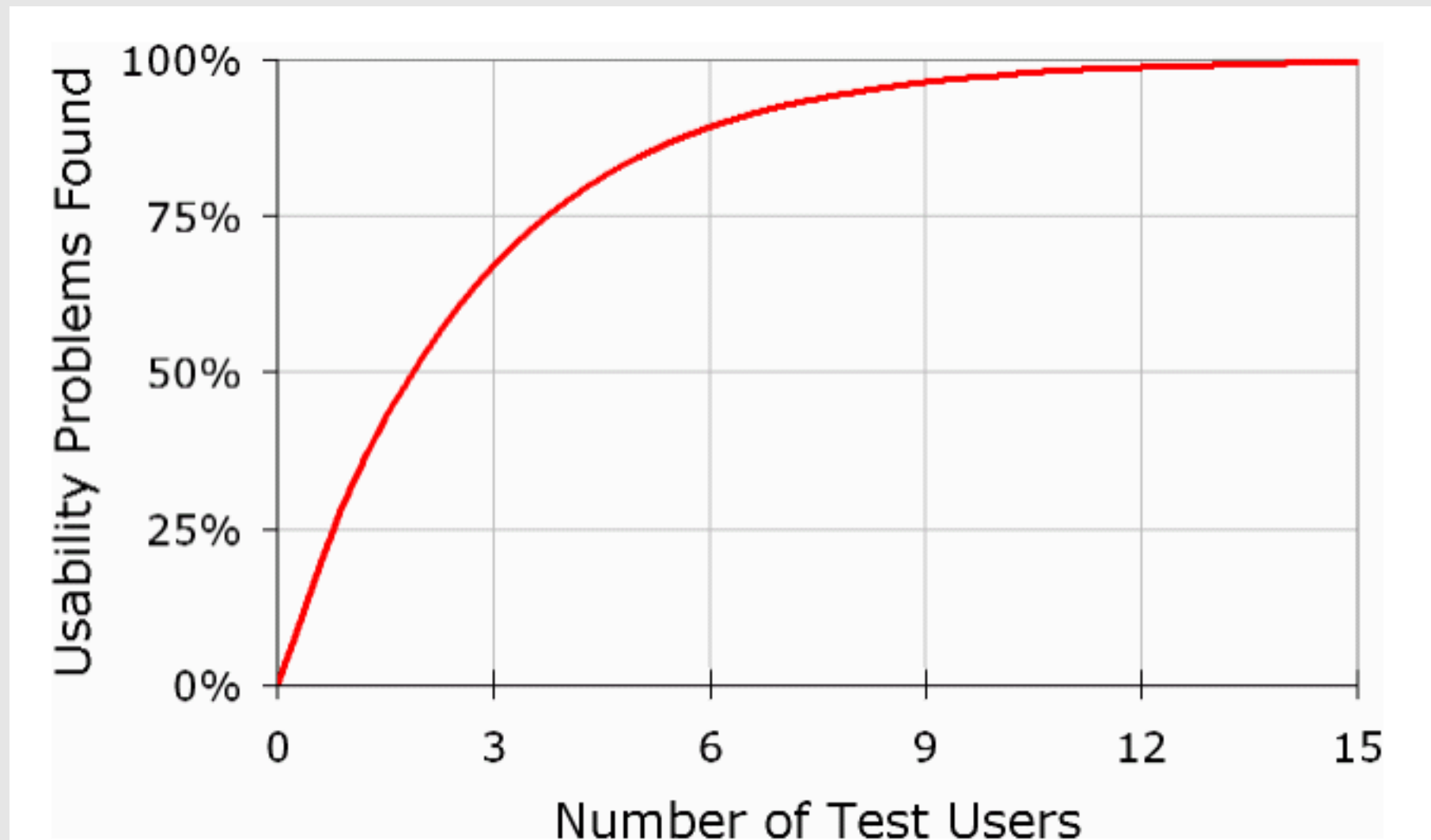
	User 1	User 2	User 3
Problem 1	X	X	
Problem 2		X	
Problem 3		X	X

Sample Size for Usability testing

Problem detection probability	Cumulative Likelihood of Detecting the Problem at Least Once (Twice)					
	0.5	0.75	0.85	0.90	0.95	0.99
0.01	68(166)	136(266)	186(332)	225(382)	289(462)	418(615)
0.05	14(33)	27(53)	37(66)	44(76)	57(91)	82(121)
0.1	7(17)	13(26)	18(33)	22(37)	28(45)	40(60)
0.15	5(11)	9(17)	12(22)	14(25)	18(29)	26(39)
0.25	3(7)	5(10)	7(13)	8(14)	11(17)	15(22)
0.5	1(3)	2(5)	3(6)	4(7)	5(8)	7(10)
0.9	1(2)	1(2)	1(3)	1(3)	2(3)	2(4)

Average Likelihood

- Avg L = 31%
- Proportion of usability problems discovered while testing with 1 user on first attempt.



	0.5	0.75	0.85	0.90	0.95	0.99
0.25	3(7)	5(10)	7(13)	8(14)	11(17)	15(22)

Neilson Group

Task Time - aka Efficiency

- Average task completion time - only record those who have completed successfully
- Mean time to failure - time users are spending on tasks before they give up or fail
- Average time on task - total time on each task
- Time saved on a task - useful for executives

Task Time - Optimal Time - KLM

- Estimate the optimal time using Keystroke Level Modelling (KLM)
- All higher level tasks can be decomposed to just these few operators:
 - Homing: moving hand to keyboard or mouse - 360ms
 - Clicking the mouse: 230ms
 - Pointing the mouse: 1,100ms
 - Mental operations (deciding): 1,350ms
- Can predict a tasks completion time by a skilled user (no errors) to within 10-20% of actual time
 - Equivalent to testing 80 users.

Task Level Satisfaction

- After completing a task ask users to rate its difficulty
 - <http://www.measureusability.com/blog/single-question.php>
 - ASQ - After Scenario Questionnaire (3)
 - NASA-TLX - Task Load Index (5)
 - SMEQ - Subjective Mental Effort Questionnaire (1)
 - UME - Usability Magnitude Estimation (1)
 - SEQ - Single Ease Question (1)

Test Level Satisfaction

The System Usability Scale (SUS) is a 10 item questionnaire with a 5 point Likert response options.

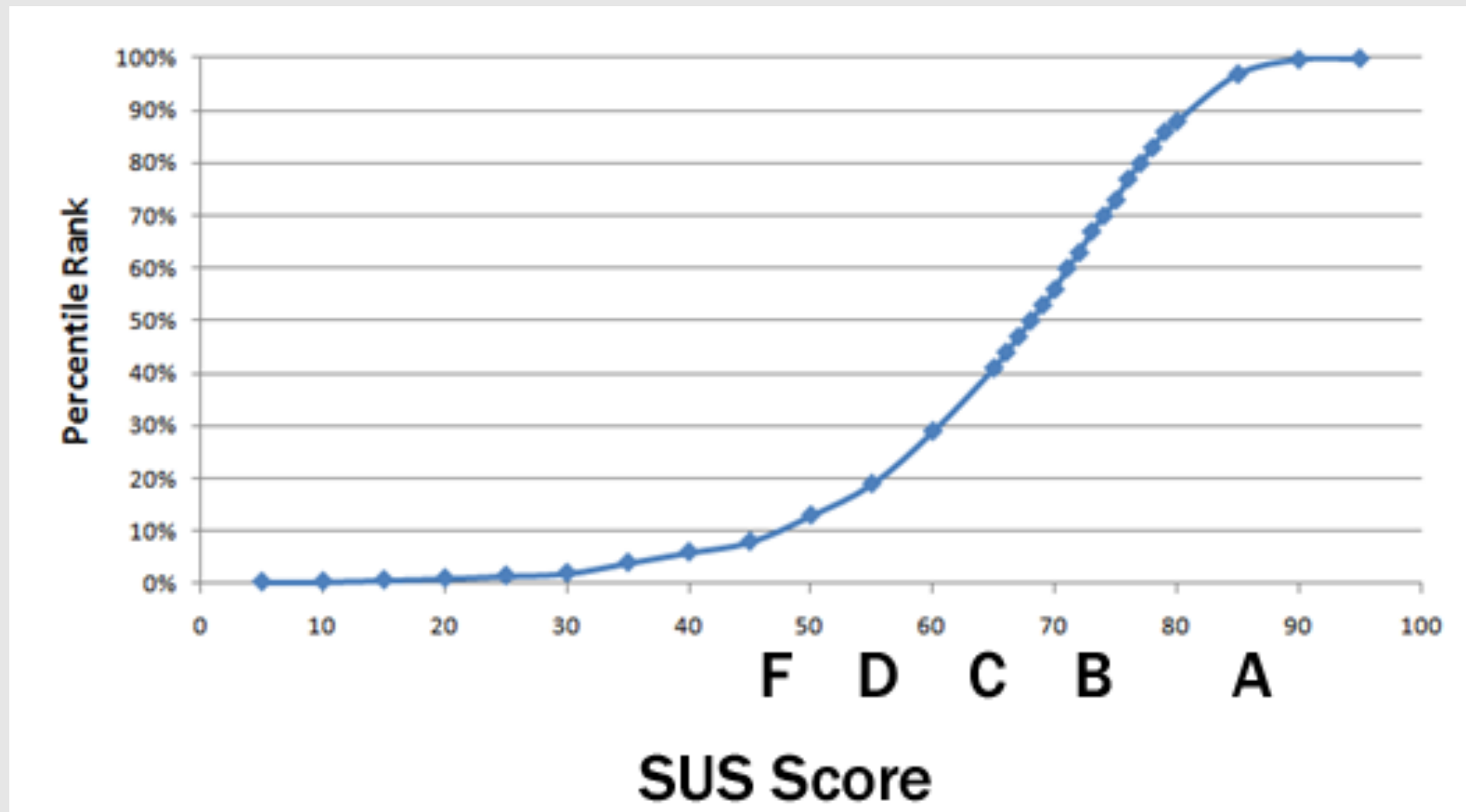
1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with the system.

System Usability Scale

Strongly Disagree	Disagree	Neither agree or disagree	Agree	Strongly Agree
1	2	3	4	5

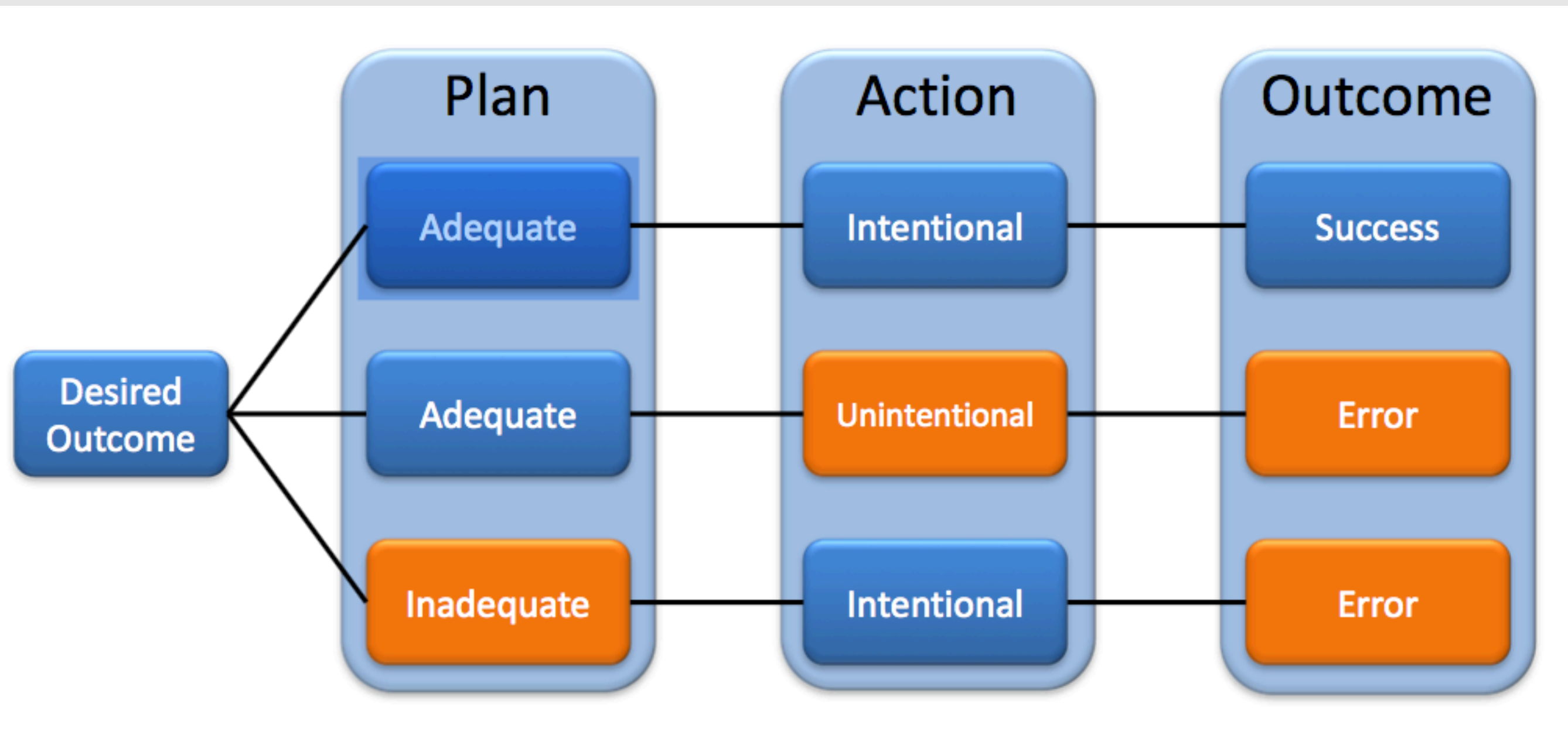
- Scoring SUS
 - For odd items: subtract 1 from the user response.
 - For even items: subtract the user response from 5.
 - Scales all values from 0-4 (4 as the most positive)
 - Add all 10 responses together and scale by 2.5
 - Converts range from 0-40 to 0-100.

System Usability Scale - Normalising



- Score > 80 is an A (top 10%)
- Score = 68 is a C (mean or average)
- Score < 51 is a F (bottom 15%)

Human Error - Plan - Action - Outcome



Human Error

- Record any unintended action, mistake, omission etc.
- E.g User types their age into the date of birth field!
- Advantages
 - Excellent in helping diagnose UI issues
- Disadvantages
 - Intensive and time consuming to collect

Expectation

- Users rate the difficulty of a task prior to attempting it
- Actual difficulty is recorded and contrasted with the expected
- Allows:
 - Tasks that were harder than expected
 - Identified for improvement
 - Tasks that were better than expected
 - Promoted

Page/Clicks Views

- Clicks are easy to measure
- Reduce the clicks = reduce the time?
 - Yes, there is some correlation
 - But, it's not strong enough
 - Clicks are not a replacement for time
 - 25% of time can be accounted for by clicks on average
- Scrolling can be activated by 1 click or many
 - Doesn't tell us much

Conversion Rate

- Special kind of completion rate - eCommerce
 - Either converted
 - or not converted
- Measures all phases of the sales process
 - Purchase
 - Checkout
 - Donate
 - Signup/register

Single Usability Measure - SUM

- A single measure of usability combining:
 - Effectiveness
 - Efficiency
 - Satisfaction
- Combines
 - Completion rates
 - Task time
 - Task level satisfaction
- Won't help in solving UI problems
- Good to benchmark against product versions or competitors

SUM Model

Figure 1: SUM Model

